

Problem

A balanced three-phase generator delivers 7.2 kW to a wye-connected load with impedance $30 - j40 \Omega$ per phase. Find the line current I_L and the line voltage V_L .

Step-by-step solution

Step 1 of 8

Refer to section 12.7 in the textbook.

Consider the load impedance of Y-connected load as,

$$\mathbf{Z}_p = 30 - j40 \Omega$$

Here,

-ve sign indicate that the load is capacitive.

Step 2 of 8

Calculate the value of phase angle.

$$\tan \theta = \frac{-40}{30}$$

$$\theta = \tan^{-1} \left(-\frac{40}{30} \right)$$

$$\theta = -53.13^\circ$$

Determine the value of power factor.

$$\text{pf} = \cos \theta$$

Substitute -53.13° for θ .

$$\begin{aligned} \text{pf} &= \cos(-53.13^\circ) \\ &= 0.6 \end{aligned}$$

Thus, the power factor is, 0.6 (leading) due to capacitive load.

Step 3 of 8

Consider the expression for the complex power $\mathbf{S}$.

$$\mathbf{S} = P + jQ$$

Here,

$$P = 7.2 \text{ kW, and,}$$

$$\begin{aligned} Q &= \left(\frac{P}{\text{pf}} \right) \sin \theta \\ &= \left(\frac{P}{\cos \theta} \right) \sin \theta \quad (\because \text{pf} = \cos \theta) \end{aligned}$$

Step 4 of 8

Determine the reactive power Q .

$$Q = \left(\frac{P}{\cos \theta} \right) \sin \theta$$

$$Q = P \tan \theta$$

Substitute 7.2 kW for P , and -53.13° for θ .

$$\begin{aligned} Q &= (7.2 \times 10^3) \tan(-53.13^\circ) \\ &= (7.2 \times 10^3)(-1.33) \\ &= -9.59 \times 10^3 \text{ VAR} \\ &\approx -9.6 \text{ kVAR} \end{aligned}$$

Thus, the value of the reactive power is $Q = -9.6 \text{ kVAR}$.

Determine the value of complex power S .

$$S = P + jQ$$

Substitute 7.2 kW for P , and -9.6 kVAR for Q .

$$\begin{aligned} S &= 7.2 \times 10^3 - j(9.6 \times 10^3) \\ &= 7.2 - j9.6 \text{ kVA} \end{aligned}$$

Here,

-ve sign shows that the load is capacitive.

Step 5 of 8

Refer to equation 12.52 in the textbook.

Consider the expression for the total complex power S .

$$S = 3I_p^2 Z_p$$

$$I_p^2 = \frac{S}{3Z_p}$$

$$I_p = \sqrt{\frac{S}{3Z_p}}$$

Here,

I_p is the phase current.

Substitute $7.2 - j9.6 \text{ kVA}$ for S , and $30 - j40 \Omega$ for Z_p .

$$\begin{aligned} I_p &= \sqrt{\frac{7.2 \times 10^3 - j(9.6 \times 10^3)}{3(30 - j40)}} \\ &= \sqrt{\frac{(7.2 - j9.6) \times 10^3}{3(3 - j4) \times 10}} \\ &= \sqrt{\frac{(7.2 - j9.6) \times 10^2}{3(3 - j4)}} \\ &= \sqrt{\frac{(12) \times 10^2}{3(5)}} \\ |I_p| &= \sqrt{80} \\ &= 8.944 \text{ A} \end{aligned}$$

Thus, the phase value of current is $I_p = 8.944 \text{ A}$.

Step 6 of 8

For a balanced $Y-Y$ system, phase currents are same as line current.

$$I_p = I_L$$

Thus,

$$I_L = 8.944 \text{ A}$$

Therefore, the required value of line current I_L is $I_L = 8.944 \text{ A}$.

Step 7 of 8

Now determine the value of apparent power S .

$$S = \frac{P}{pf}$$

Here, P is the real power.

Substitute 7.2 kW for P , 0.6 for pf .

$$\begin{aligned} S &= \frac{72 \times 10^3}{0.6} \\ &= 120 \text{ kVAR} \end{aligned}$$

Step 8 of 8

Determine the value of the load voltage V_L .

$$S = \sqrt{3} V_L I_L$$

$$V_L = \frac{S}{\sqrt{3} I_L}$$

Substitute 120 kVAR for S , and 8.944 A for I_L .

$$\begin{aligned} V_L &= \frac{(120 \times 10^3)}{\sqrt{3} (8.944)} \\ &= 7.746 \times 10^3 \text{ V} \\ &= 7.75 \text{ kV} \end{aligned}$$

Therefore, the required value of line voltage V_L is 7.75 kV .